

● NATIONAL LEVEL HACKATHON 2026

ArogyaGram

Rural India's AI-Powered Emergency Health Response Network

When the nearest hospital is 40 km away and the ambulance takes
90 minutes,
ArogyaGram responds in seconds.



Gemini AI
Triage



SOS
Network



Verified
Volunteers



Med
Adherence

THE CRISIS

Rural India's Healthcare Desert

65%

of India lives in rural areas
with <30% of healthcare
infrastructure

90+

minutes average ambulance
response time in Tier-3
districts

1:10K

doctor-to-patient ratio in
rural Kerala vs 1:1000 urban
average

The Daily Reality

- Elderly diabetic patients missing insulin doses because the pharmacy is 12 km away
- Falls and cardiac events going unreported for hours in isolated homesteads
- ASHA workers unable to escalate urgent cases without real-time clinical guidance
- Language barriers preventing patients from describing symptoms correctly

THE SOLUTION

ArogyaGram: One Platform, Four Critical Pillars



AI Clinical Triage

Gemini-powered multilingual symptom assessment in Malayalam, Hindi, Tamil & English. Provides risk classification, first-aid guidance, and escalation decisions — no doctor needed at Step 1.



Deterministic SOS

Keyword-triggered emergency protocols that bypass AI and directly call 9539141210 (configured rescue number) + display geo-tagged location. Falls, chest pain, unconsciousness → instant response.



Medication Adherence

Smart structured med tracking with food-timing awareness, native time pickers, and daily adherence scores. Reduces the #1 cause of rural chronic illness complications — missed doses.



Volunteer Task Network

Community-verified neighbors accept geo-matched tasks for medicine pickup, clinic transport & welfare checks. AI-calculated match scores ensure the right person responds to every request.

INTELLIGENT ORCHESTRATION

From Distress Signal to Coordinated Response

1

Voice / Text Intake

Patient speaks or types in their native language. Natural language processing understands context, urgency, and category — not just keywords.

2

AI Situation Engine Triage

Gemini analyzes the situation against clinical heuristics. Emergency keywords → immediate SOS. Non-emergency → structured care case with risk classification (LOW / MEDIUM / HIGH).

3

Volunteer Matching & Dispatch

MatchingService scores nearby verified volunteers by proximity, skills, vehicle availability, and past rating. Best match receives push notification with task details.

4

Tracked Resolution & Feedback

Unified Health Timeline records all events. Volunteer marks task complete. Patient rates experience. Data feeds back to improve future matching quality.

TECHNOLOGY

Gemini-Powered Intelligence with Deterministic Safety

AI Capabilities

- ✓ Multilingual symptom parsing (5 languages)
- ✓ Risk stratification: LOW / MEDIUM / HIGH / CRITICAL
- ✓ Intent classification: pharmacy, transport, grocery, welfare
- ✓ Situation synthesis with structured care plans
- ✓ User-configurable API key (zero vendor lock-in)

Safety Guardrails

- ⚡ AI NEVER replaces emergency services
- ⚡ Critical keywords bypass AI → direct SOS dial
- ⚡ Volunteers clearly scoped: no clinical decisions
- ⚡ All AI responses include medical disclaimer

Tech Stack

Frontend	React + TypeScript + Vite
UI System	Tailwind CSS (Dark Mode)
AI Engine	Google Gemini 2.5 Flash
Maps	Leaflet + OpenStreetMap
Backend	Supabase (PostgreSQL + RLS)
Distribution	PWA + Android APK (Capacitor)
Voice	Web Speech API (en-IN)

✓ 2065 modules · 1.35s build · 7/7 tests passing

COMMUNITY + EMERGENCY

Neighbors as First Responders, Technology as the Backbone

Volunteer Network

8 km

configurable service radius per volunteer

100pt

AI match scoring (proximity + skill + rating)

Maps

Live Leaflet map view of all nearby requests

Verified

Community-endorsed identity before onboarding

Tasks are strictly scoped — medicine pickup, transport, companionship. Volunteers are *not* healthcare providers. Gemini AI generates a Safety Guardrail reminder on every active mission.

SOS System

One-tap Dial

Directly calls rescue number 9539141210

GPS Location

Geolocation shared with responders

Emergency Contacts

WhatsApp alert to pre-set family numbers



SOS RESCUE

IMPACT & TRACTION

Built for Scale, Designed for Dignity

5

Indian languages supported at launch

4

core health pillars in a single unified app

<5s

AI triage response time per case

Who It Helps

- Elderly & chronically ill rural citizens
- ASHA workers and frontline health staff
- Families of patients living far from facilities
- Community volunteers wanting to contribute
- PHC doctors managing case overflow

Scalability

- PWA + Android APK — offline-capable
- Supabase backend scales to millions of rows
- Any district in India can be onboarded
- User-supplied Gemini API keys — zero hosting cost
- Modular: pluggable for govt ABDM integration

Pilot target: **Kanjirappally block, Kottayam district, Kerala**
Population: 220,000 · 4 PHCs · 12 sub-centers · 180 ASHA workers



No Rural Citizen Should Face Illness Alone.

ArogyaGram isn't just an app — it's a pledge that technology can bridge India's most persistent health equity gap.

The Ask

Seed partnership with NHM Kerala for district-level pilot deployment

Next Steps

ASHA worker beta group, Supabase production backend, offline PWA caching

Try Now

Live APK available · GitHub: [jvstin47/ArogyaGram-3](#)

ArogyaGram

Because every second counts. Especially 40 km from the nearest hospital.